

- 2 (a) speed (of object) at surface (of planet) / specified starting point so that object may move to infinity / escape gravitational field of planet B1
- (b) loss in kinetic energy = gain in (gravitational) potential energy C1
- $$\frac{1}{2}mv^2 = \frac{GMm}{R}$$
- But $g = \frac{GM}{R^2}$ C1
- Hence $v = \sqrt{2gR}$ (no mark for answer)
- assumption: e.g. planet is isolated / no friction / no atmosphere etc. A1
- (c) $v = \sqrt{2(1.6)1.7 \times 10^6}$ C1
- $= 2.3 \times 10^3 \text{ m s}^{-1}$ A1
- (d) $\frac{1}{2}(4 \times 1.66 \times 10^{-27})(2.3 \times 10^3)^2 = \frac{3}{2}(1.38 \times 10^{-23})T$ C1
- $T = 850 \text{ K}$ A1
- (e) atoms have a distribution of speeds / atoms may collide in upper atmosphere and gain speed B1

- 3 (a) (i) Initial kinetic energy of block = $\frac{1}{2}mv^2 = \frac{1}{2}(1.5)(1.5)^2 = 1.69 \text{ J}$ B1